Aleksei Pshenichnyi

AI/ML Research Engineer · Generative Video · Technical R&D Lead

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Summary

Hands-on AI/ML engineer with 15+ years in software, specializing in generative video and audio-driven facial animation. Co-founder of Elai, an AI video startup acquired by Panopto in 2024. Works end to end, from model research and evaluation to production deployment and real-time inference, with a background in distributed systems. Authorized to work in the EU.

Skills

AI / ML: PyTorch, Transformers, Convolutional Autoencoders, Generative Models, Computer Vision, LLMs, Multimodal Models

Specializations: Audio-Driven Facial Animation, Lip Sync, Neural Video Generation, Model Training, Evaluation Frameworks

Performance & Systems: Python, CUDA, Cython, Numba, JIT, Streaming Inference, Distributed Systems

Data Engineering: Kafka, Cassandra, Greenplum, PostgreSQL, Scala, Java, Akka

Leadership: R&D Team Leadership, Research Planning, Technical Architecture

Experience

Panopto, Head of AI Research — Nov 2024 – present

Joined through the acquisition of Elai. Lead avatar video generation R&D and new generative AI features; about 70% hands-on.

- Released a new generation of audio-driven facial animation models to production; users generated 300K+ avatar videos in 2024-2026.
- Fixed over-smoothed articulation in audio-to-animation models through training data design and architecture changes, now in production.
- Shipped a feature that generates animated charts from text prompts or slides, using LLMs for structured chart descriptions and multimodal models to read slides.
- Lead an R&D team of 4-5 engineers.

Elai, Co-Founder, Chief R&D Officer (Hands-on Principal Engineer) — May 2021 – Dec 2024

AI video generation startup, acquired by Panopto in October 2024.

- Built production neural systems for audio-driven facial animation and talking-head video generation with transformer-based models and convolutional autoencoders in PyTorch.
- Led a 4-5-person R&D team while developing models and running experiments myself.
- Set up training, experiment tracking and evaluation combining automated metrics with structured human assessments and preference tests.
- Built a streaming real-time avatar MVP (LLM, TTS, animation, neural rendering) at 25+ FPS, with 2-3 seconds from first audio chunk to first frame.
- Optimized critical paths with CUDA, Cython, Numba and JIT compilation.

Eidolon, Head of Analytics (part-time) — Jan 2021 – Feb 2022

Designed and led development of an event analytics platform for mobile games, covering data models, processing and reporting.

DataSpike, Technical Lead (Co-Founder) — Oct 2019 – Jan 2021

Led R&D of an early-stage NLP product for AML and KYC in financial institutions (Scala, Python, SPARQL).

Quadcode (IQ Option), Software Development (Data Engineering) Team Lead — May 2017 – Oct 2019

Architect and hands-on lead of the analytics and ML data platform for the IQ Option fintech product.

- Built a distributed event analytics platform with reliable delivery, deduplication and consistency checks; this platform and the later one at Eidolon handled thousands of events per second (peaks of tens of thousands) on tens of TB of hot data.
- Cut recurring reports from tens of minutes to seconds with pre-aggregation, partitioning and query plan optimization.

- Led ML systems on the platform, covering paying-user prediction for ad spend, fraud detection and trading news recommendations.
- Built real-time analytics infrastructure with Scala and Akka and pipelines with Python, Scala and Postgres.

Ice Owl, Software Developer (Backend & Game AI) Oct 2010 – Apr 2017

Server-side game logic, multiplayer features and AI for social and strategy games in Scala and Java.

- Designed memory-efficient planning for autonomous game agents, cutting precomputed data from an estimated tens of GB to about 100 MB per map and running around 1,000 agents across hundreds of concurrent sessions on one server.

Education

Computer Science Center, Data Analysis Saint Petersburg, Russia

- Evening postgraduate program (non-degree) Sept 2014 – June 2017

Siberian Federal University, Computer Science Krasnoyarsk, Russia

- Specialist degree (5-year, MSc equivalent) Sept 2004 – June 2009

Languages

English (Upper-intermediate (B2)) · Russian (Native) · Spanish (Basic (A1))